Prelims Introductory Calculus MT 2026: Sheet 2

1. Find solutions of the following ODE problems including their domains:

   (a)
   $$\frac{\mathrm{d}y}{\mathrm{d}x} + xy = x, \quad y(0) = 0;$$

   (b)
   $$2x^3\frac{\mathrm{d}y}{\mathrm{d}x} - 3x^2y = 1, \quad y(1) = 0;$$

   (c)
   $$\frac{\mathrm{d}y}{\mathrm{d}x} - y\tan x = 1, \quad y(0) = 1.$$

2. Solve the following differential equations:

   (a) $(1-x^2)\dfrac{\mathrm{d}y}{\mathrm{d}x} + 2xy = (1-x^2)^{3/2}$, (b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} - (\cot x)y + \csc x = 0.$

3. By treating y as the independent variable, or otherwise, find the implicit solution and the domain of $y(x)$ where
   $$(x + y^3)\frac{\mathrm{d}y}{\mathrm{d}x} = y, \qquad y(1) = 2.$$

4. Find the general solutions of the following homogeneous linear equations:

   (a) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} - y = 0;$ (b) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0;$ (c) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 3\dfrac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0;$

   (d) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\dfrac{\mathrm{d}y}{\mathrm{d}x} + 3y = 0;$ (e) $\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3} + \dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + \dfrac{\mathrm{d}y}{\mathrm{d}x} + y = 0;$ (f) $\dfrac{\mathrm{d}^4y}{\mathrm{d}x^4} + \dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0.$

5. By using the substitution $z = \ln x$, or otherwise, find the solution of the boundary value problem
   $$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0 \qquad y(1) = 0, \quad y(e) = 1.$$

6. Find the general solutions of the following inhomogeneous equations:

   (a)
   $$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 4e^{3x} + 2\sin x;$$

   (b)
   $$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 3y = x^2 + 2.$$